

MONAHANS 6 ENE, TX, USA

WMO: 747290

Lat: 31.6219N

Lon: 102.8071W

Elev: 2724

StdP: 13.31

Time Zone: -6.00 (NAC)

Period: 03-19

WBAN: 03047

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.8	25.6	2.4	6.9	45.1	5.9	8.2	49.3	20.2	57.3	16.8	59.4	3.9	N/A	0.412

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.7	102.6	66.3	100.1	66.9	97.8	67.0	72.7	88.4	71.8	87.8	71.1	87.2	6.6	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.2	118.8	76.0	68.1	114.1	75.8	67.0	110.0	75.9	38.1	88.2	37.3	87.8	36.7	87.1	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.5	13.3	11.9	DB	14.3	106.9	4.3	2.5	11.2	108.7	8.7	110.2	6.3	111.6	3.2	113.4	(4)
			WB	12.5	74.4	4.2	1.0	9.4	75.1	7.0	75.7	4.6	76.3	1.5	77.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.9	46.0	50.7	59.3	66.6	74.3	82.6	83.1	82.7	76.0	66.7	55.0	46.8
	DBStd	15.42	8.49	9.80	9.05	7.79	7.50	4.82	4.18	4.31	5.53	8.09	8.80	8.43
	HDD50	526	177	101	27	3	0	0	0	0	0	5	52	161
	HDD65	2254	588	404	216	73	18	0	0	0	3	77	311	564
	CDD50	6332	54	121	317	502	754	979	1025	1015	780	521	202	62
	CDD65	2586	0	4	41	122	307	529	560	550	333	129	11	0
	CDH74	31923	37	179	757	1767	3856	6725	6927	6667	3352	1383	229	44
	CDH80	16902	2	36	239	770	2053	3919	3942	3780	1594	521	44	2
Wind	WSAvg	5.7	5.2	5.9	6.2	6.4	6.5	6.8	6.1	5.3	5.2	5.0	4.8	4.9
Precipitation	PrecAvg	14.2	0.6	0.7	0.7	0.6	1.4	1.3	1.6	1.9	1.9	1.3	0.6	0.7
	PrecMax	24.8	1.9	1.7	2.6	2.6	7.0	3.7	5.9	4.2	5.3	4.6	4.1	3.1
	PrecMin	8.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	4.8	0.6	0.6	0.7	0.6	1.7	1.1	1.4	1.1	1.5	1.4	0.9	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.3	84.9	90.3	95.2	103.5	106.8	103.8	104.0	99.4	93.1	85.3	79.1
		MCWB	51.8	54.6	55.9	57.6	61.1	63.9	67.8	68.5	65.7	62.5	57.4	52.8
	2%	DB	73.5	79.3	86.2	92.0	98.7	103.4	100.7	100.9	95.4	89.4	80.1	73.8
		MCWB	50.0	53.1	55.1	57.2	60.1	64.5	68.3	68.2	66.3	62.7	56.7	51.4
	5%	DB	68.2	75.0	82.3	88.7	95.3	100.0	98.5	98.4	92.7	86.2	76.2	68.3
		MCWB	47.7	51.3	54.4	56.8	60.0	66.5	68.2	68.3	66.4	62.1	55.4	49.4
	10%	DB	62.9	69.6	78.2	84.6	91.4	96.8	95.9	95.7	89.2	82.2	71.7	63.0
		MCWB	45.7	49.3	53.6	56.6	61.1	67.4	68.3	68.6	66.2	61.7	53.6	47.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.1	59.9	63.7	66.3	71.4	73.5	74.7	73.3	72.8	71.3	64.5	57.5
		MCDB	68.9	71.8	77.8	79.1	85.1	90.3	86.4	89.8	86.7	84.9	72.5	65.7
	2%	WB	51.7	57.0	61.6	64.3	70.0	72.3	72.5	72.1	71.4	69.1	61.2	54.9
		MCDB	70.0	68.6	73.1	75.9	82.4	89.1	87.5	89.3	85.2	79.8	68.5	63.8
	5%	WB	49.2	54.2	59.5	62.6	68.7	71.3	71.5	71.2	70.5	67.2	59.1	52.5
		MCDB	65.7	69.6	71.3	74.5	80.5	88.0	87.5	88.8	83.8	77.4	68.8	64.3
	10%	WB	46.6	51.8	57.2	60.9	67.3	70.4	70.8	70.4	69.5	65.3	56.6	49.8
		MCDB	61.8	68.6	69.0	73.5	78.9	87.0	86.9	88.2	82.1	76.1	68.1	62.9
Mean Daily Temperature Range	5% DB	MDBR	27.1	28.2	28.9	29.2	27.4	26.2	23.7	24.0	22.8	25.7	27.2	26.7
		MCDBR	36.8	36.4	35.3	34.7	33.7	30.6	27.4	27.3	28.1	30.4	32.6	35.6
		MCWBR	17.5	16.1	13.2	12.6	12.4	8.4	7.2	7.2	8.5	10.8	14.1	17.2
	5% WB	MCDBR	33.2	31.4	28.6	27.9	26.0	26.0	23.2	24.1	21.0	22.8	24.9	28.8
		MCWBR	16.3	15.1	11.4	11.8	11.6	7.7	6.4	6.8	6.8	9.3	12.9	15.5
Clear-Sky Solar Irradiance	taub	0.263	0.272	0.291	0.326	0.349	0.377	0.399	0.371	0.353	0.307	0.284	0.268	
	taud	2.583	2.561	2.516	2.402	2.397	2.374	2.351	2.413	2.432	2.555	2.548	2.570	
	Ebn at Noon	311	317	316	305	297	287	280	288	290	299	299	304	
	Edh at Noon	25	29	32	37	38	39	40	37	35	28	26	24	
All-Sky Solar Radiation	RadAvg	1116	1390	1794	2181	2323	2413	2292	2127	1791	1477	1174	1006	
	RadStd	104	139	124	133	143	108	116	93	115	150	116	88	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	+1.84	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page